SWJTU ////////

Progress Report

答辩学生：杂尔 Muhammad Abuzar Sajjad (2025330026)

指导教师：马磊 Ma Lei



西南交通大學
Southwest Jiaotong University



西南交通大學
Southwest Jiaotong University

Transfer Engine for Zero-Shot Robot Deployment

Introducing the Neuro-Mathematical Transfer Engine (NMTE)

Recap: Where We Were

- Previous Approach: Muscle Memory Adaptation
- Method:
  Use VLMs to build a digital twin, then use Residual Reinforcement Learning (RL) to "warp" the old trajectory to fit the new room.
- The Goal:
  Avoid retraining AI from scratch every time a robot moves to a new workstation.

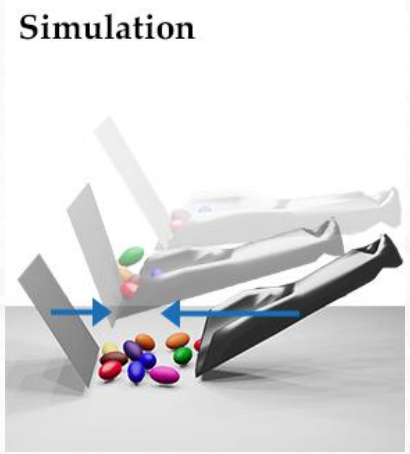
The Reality Check: Current Problem

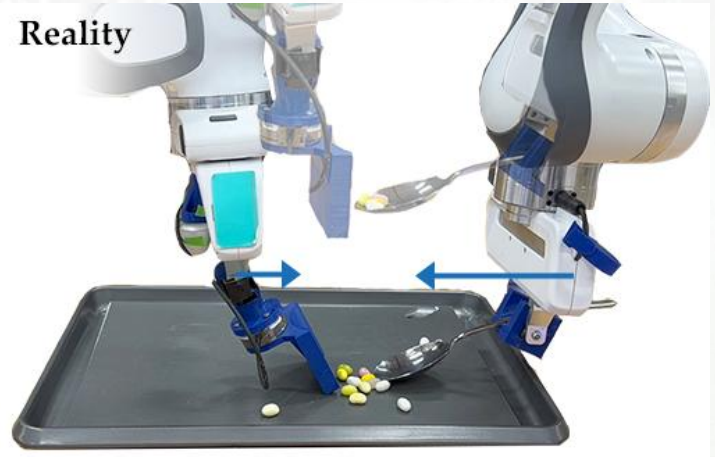
西南交通大学 Southwest Jiaotong University

Problem:

Using models like Sam3D or SayCan to scan a room, and creating a digital twin, and then using Residual RL to tweak the robot's existing 'muscle memory' to fit the new layout works beautifully for basic navigation and static obstacles.

However, when we looked at deploying this to actual physical hardware in messy industrial environments like railways or high-speed warehouses, significant gaps in the literature were found.





The Reality Check: Identifying Critical Gaps

西南交通大学 Southwest Jiaotong University

- Gap 1 & 2:
  Unpredictable Physics. Pure RL fails when friction, heavy loads, or loose terrain change unexpectedly (The "Reality Gap").
- Gap 3:
  The "Siloed Body" Problem. A policy trained for a robotic arm cannot be transferred to a robot dog.
- Gap 4:
  No Safety Guarantees. RL is a "black box." If the robot encounters an unseen scenario, it might crash. We cannot formally guarantee it won't break itself or hurt a human.
- Gap 5:
  Edge Compute Limits. Running heavy AI models continuously drains battery and overloads the robot's onboard computer.

The Pivot: Math-Guided Neural Networks (NMTE)

西南交通大学 Southwest Jiaotong University

Proposed Solution:

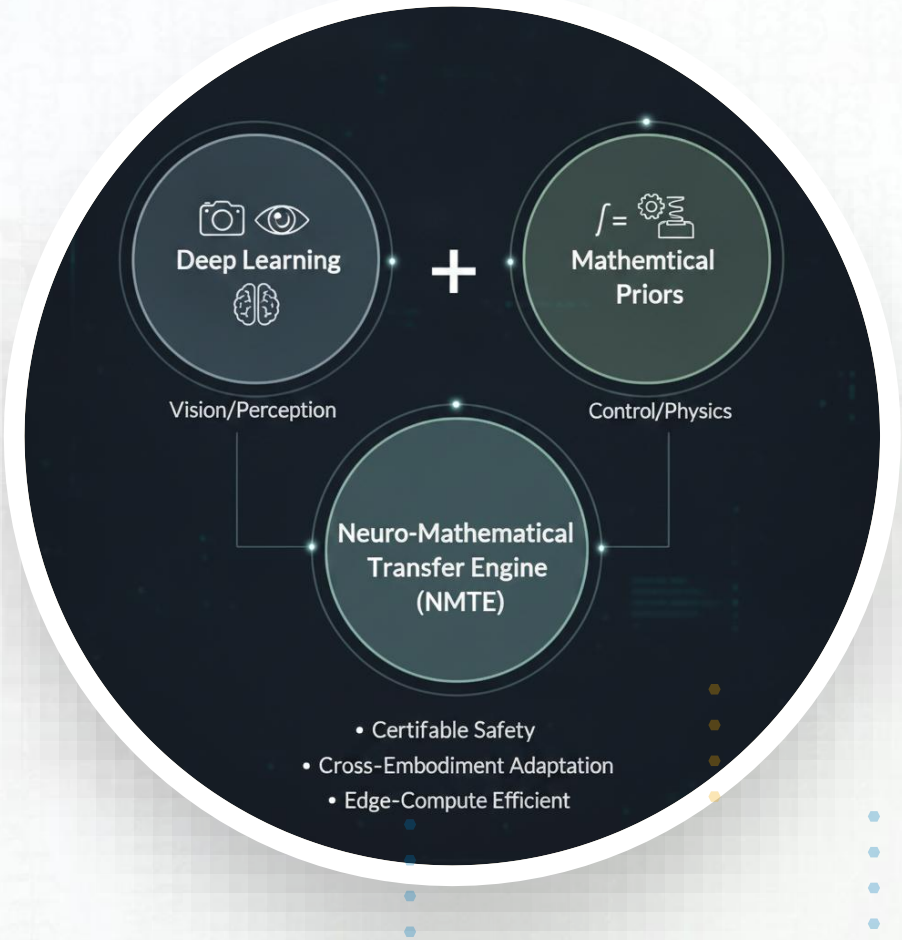
The Neuro-Mathematical Transfer Engine (NMTE)

The Concept:

Keep Deep Learning for seeing and understanding the environment, but use strict Mathematical Control Theory to execute the physical movements.

Benefits:

- Certifiable safety
- Instant adaptation to new robot bodies (cross-embodiment)
- Computationally lightweight on the edge



Beating Unpredictable Physics: Koopman Operator Theory

西南交通大学 Southwest Jiaotong University

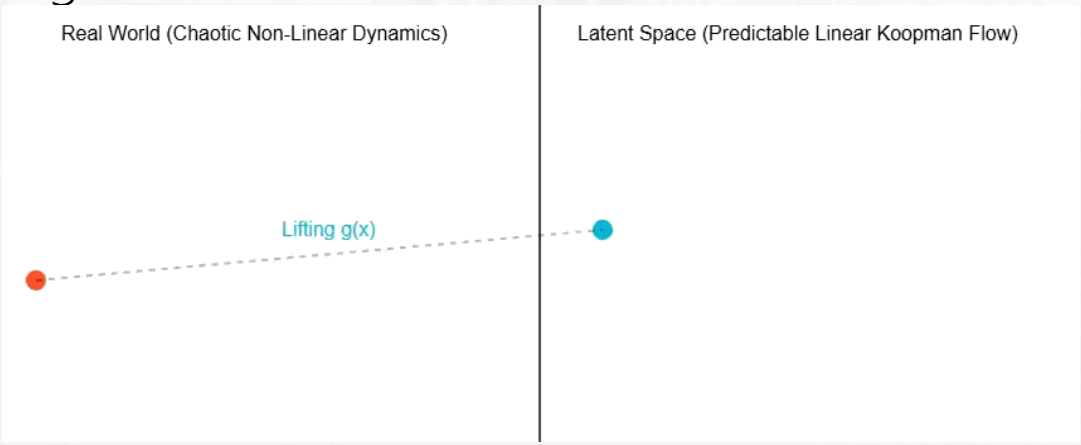
The Gap:

Real-world friction, sliding, and terrain invalidate simulation training.

The Math:

Koopman Operator Theory

How it works: Non-linear, chaotic physics (like slipping on loose gravel) are incredibly hard for an AI to predict. Koopman theory maps these complex physics into a higher "dimension" where they behave as simple, straight lines.



The Result:

The robot instantly absorbs real-world contact variations without needing to recalculate the whole AI policy.

Solving the Body Problem: Optimal Transport

西南交通大学 Southwest Jiaotong University

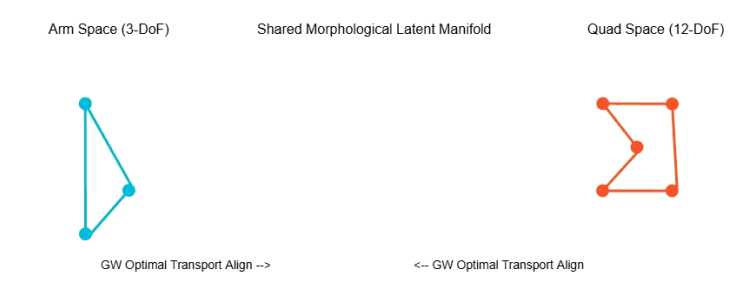
The Gap:

Transferring skills across different robot shapes (Cross-Embodiment).

The Math:

Gromov-Wasserstein Optimal Transport

How it works: Acts as a "Universal Translator." It looks at the topological structure of a task rather than the exact joint coordinates.



The Result:

Teach a robotic arm how to pick up a box, and mathematically project that exact skill onto a robot dog, zero-shot.

Guaranteeing Safety: Control Barrier Functions (CBF)

西南交通大學 Southwest Jiaotong University

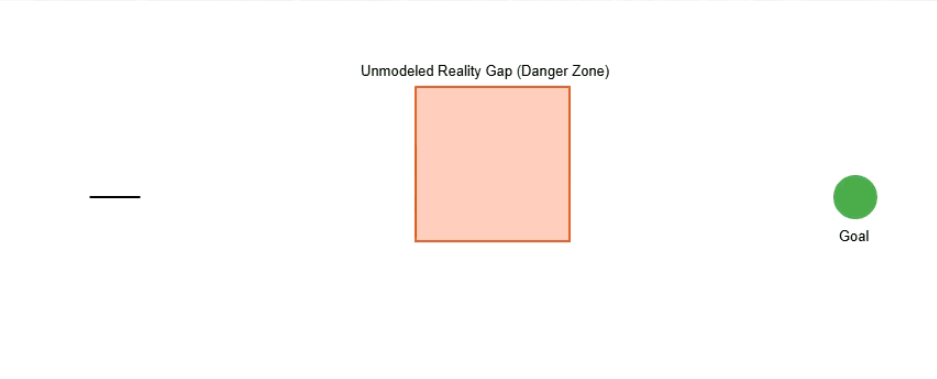
The Gap:

Neural networks cannot guarantee collision avoidance during sim-to-real shifts.

The Math:

Disturbance-Observer Control Barrier Functions (DOB-CBF)

How it works: An absolute mathematical "override switch." The AI suggests a movement. The CBF checks the math. If the move will cause a crash or exceed motor limits, the CBF actively overrides the AI and redirects the robot to a safe zone.



The Result:

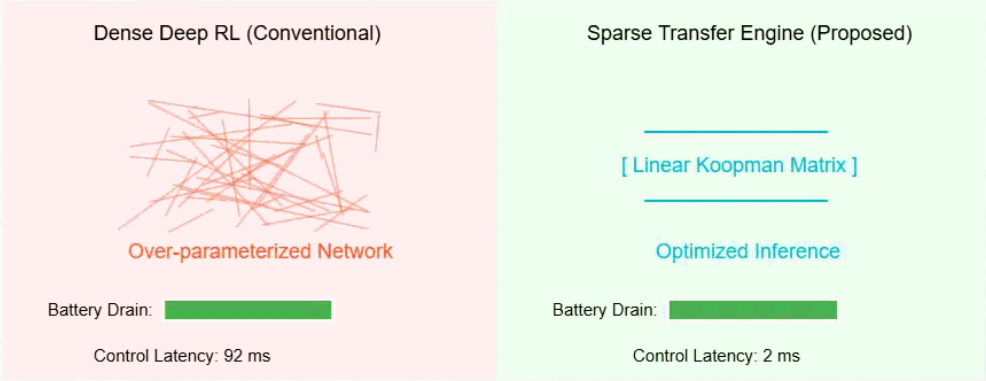
100% mathematically certifiable safety in shared human-robot workspaces.

Energy and Compute Efficiency (Edge Deployment)

西南交通大学 Southwest Jiaotong University

The Gap:

Complex AI models drain battery and cause latency.



The NMTE Advantage:

- Heavy lifting (training the Optimal Transport and Koopman matrices) happens offline in the simulation.
- Real-world execution only requires solving simple linear math and lightweight quadratic programs.

The Result:

High-speed (>500 Hz) control loops running purely on edge devices with minimal power drain.

Updated Research Roadmap

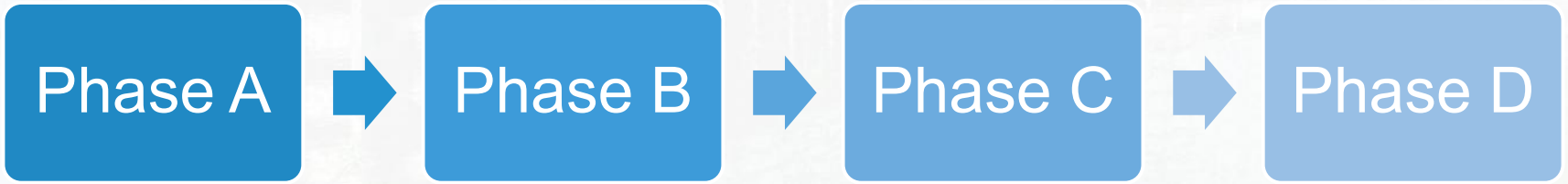
西南交通大學 Southwest Jiaotong University

Phase A: Dynamic Modeling
Validate Koopman functions for task-invariant flow fields.

Phase B: Latent Alignment
Formulate zero-shot cross-embodiment transfers (Arm to Quadruped).

Phase C: Safety Certification
Integrate real-time CBFs for collision avoidance.

Phase D: Hardware Validation
Deploy the NMTE on physical ROS2 edge-compute modules.



谢谢观看

Thank You!

答辩学生: Muhammad Abuzar Sajjad